

1 Nodes

The idea of nodes is that each system within a robot is a node in and of itself; with nodes being able to send and receive messages from other nodes. To find a list of active nodes, we use the command:

```
$ros2 node list
```

So if we run the following command to open turtlesim, we get the following output:

Input:

```
$ros2 run turtlesim turtlesim_node  
$ros2 node list
```

Output:

```
/turtlesim
```

1.1 Remapping

We can change values that each node possesses via remapping commands. For example, the following command changes our turtlesim node's name:

```
$ros2 run turtlesim turtlesim_node --ros-args --remap__node:=my_turtle
```

In this case, we run the turtlesim once more, creating a new sim instance; however, the node is now named my_turtle.

2 Topics

The fundamental idea behind topics is the publisher-subscriber model. The publishers are the nodes who send the message and the subscribers receive it. Multiple publishers and subscribers can connect to one topic at the same time.

A message sent from one publisher is transmitted to all subscribers.